

Generation Interconnection System Impact Study Report

For Queue Project Q20-1593

Hat Rapids 138 kV

May 23, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1593. The Interconnection Customer proposes a 90.6 MW solar photovoltaic facility interconnecting at the Hat Rapids 138 kV substation in the Virginia area. The request was received on October 10, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 469 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,295,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Paokee-Zephyrhills 138 KV
Contingency: Cady-Easton 2 345 KV
Loading: 120.6% (Rating: 231 MVA)
DFAX: 8.0%
Pre-project loading: 117.5%
2. Facility: St Clair-Pahokee 138 KV L
Contingency: Cady-Easton 2 345 KV line
Loading: 113.7% (Rating: 225 MVA)
DFAX: 8.0%
Pre-project loading: 110.5%
3. Facility: Westcoast Cellufibre-Leavitt 138 KV Circuit 1
Contingency: Westcoast Cellufibre-Leavitt 138 KV Ckt 2
Loading: 105.6% (Rating: 384 MVA)
DFAX: 5.7%
Pre-project loading: 104.3%
4. Facility: Leavitt-Westcoast Cellufibre 138 KV L
Contingency: Westcoast Cellufibre-Leavitt 138 KV Ckt 1
Loading: 105.6% (Rating: 384 MVA)
DFAX: 5.7%
Pre-project loading: 104.3%
5. Facility: Chancellor Row-Bear Island 345 KV
Contingency: Leavitt-Birchwood 345 KV line
Loading: 100.3% (Rating: 2,328 MVA)
DFAX: 17.7%
Pre-project loading: 99.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Paokee-Zephyrhills 138 KV	Rebuild line with higher-capacity conductor	\$4,933,000	10% (\$485,000)
St Clair-Pahokee 138 KV L	Rebuild line with higher-capacity conductor	\$16,944,000	10% (\$1,664,000)
Leavitt-Westcoast Cellufibre 138 KV	Reconductor line	\$430,000	0% (\$2,000)
Westcoast Cellufibre-Leavitt 138 KV Cir. 2	Reconductor line	\$545,000	0% (\$3,000)
Chancellor Row-Bear Island 345 KV	Reconductor line	\$17,651,000	1% (\$141,000)
	Total allocated to project		\$2,295,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.